

Comprehending ER diagrams and UML class diagrams

TASK: T1

TREATEMENT: CD_EX

First Name:.....

Last Name:.....

ID:.....

1. Comprehending

Start Time:

Figure 1 shows the UML class diagram of a system managing the hospital activities. In particular, the diagram regards the patients, doctors, examinations, and laboratory information.

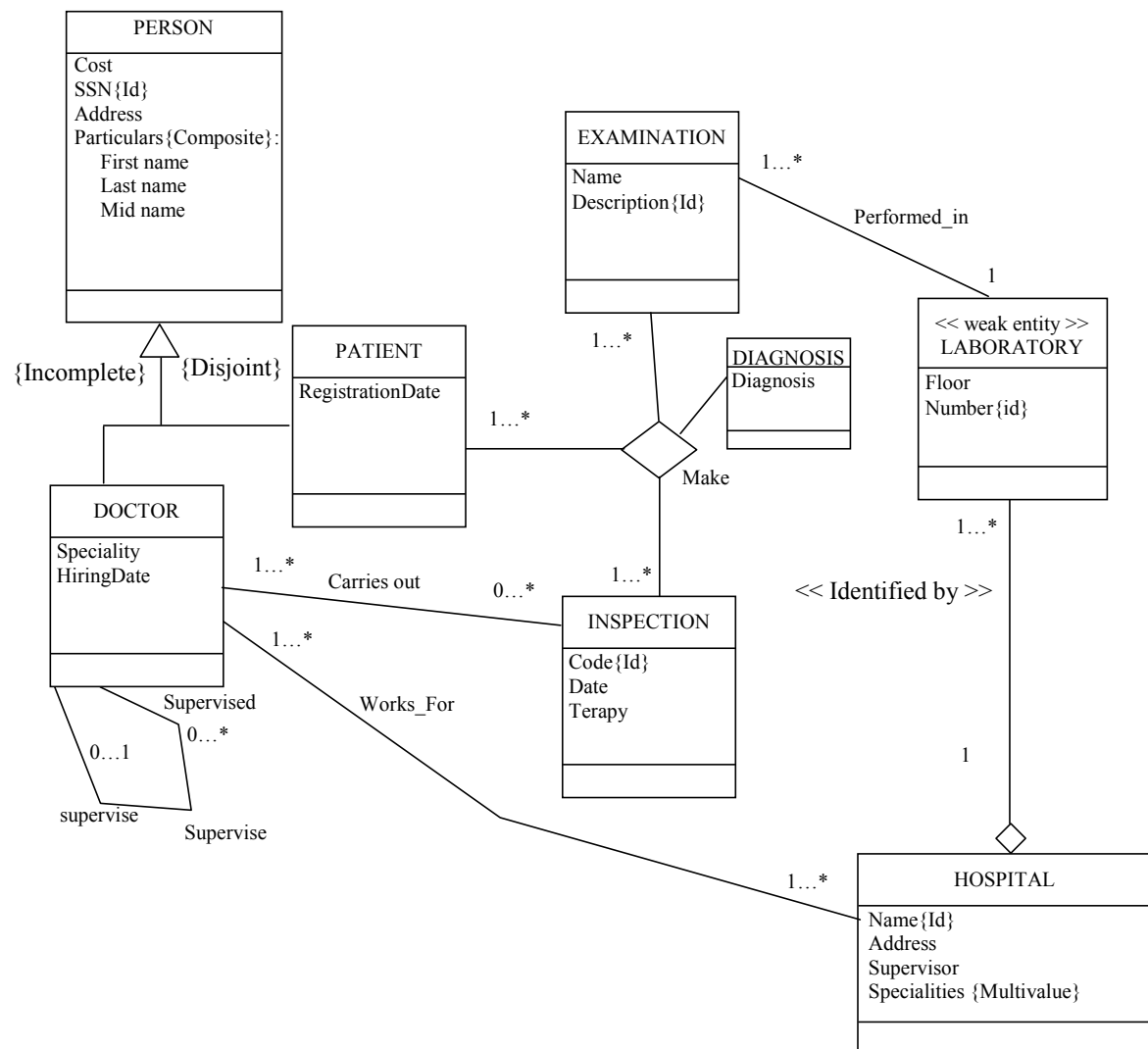


Figura 1: UML class diagram of a system managing hospital information

Analyze the diagram in Figure 1 and answer the following questions.

Q1 Focusing on class Patient, which of the following entries is true?

- ☐ Given a patient, it is possible to obtain his/her registration date.
- ☐ Given a patient, it is possible to obtain his/her birth date
- ☐ Given a patient, it is possible to determine the days he/she was in the hospital.

Q2 Focusing on class Examination, which of the following entries is true?:

- ☐ An examination can be uniquely identified by using its name.
- ☐ An examination can be uniquely identified by using its description and name
- ☐ An examination can be uniquely identified by using its description

Q3 Focusing on class Patient, which of the following entries is true?:

- ☐ Given a patient, it is possible to obtain the initial of his/her name
- ☐ Given a patient, it is possible to obtain his/her unmarried last name
- ☐ Given a patient, it is possible to obtain his/her last name

Q4 Focusing on classes Doctor and Hospital, which of the following entries is true?:

- ☐ A hospital has a unique speciality
- ☐ A doctor has a unique speciality
- ☐ A hospital may have several specialities

Q5 Focusing on class Doctor and its relationships, which of the following entries is true?:

- ☐ A doctor may have no supervisor
- ☐ A doctor may have no supervisor may be supervisor of several doctors
- ☐ A doctor has several supervisor

Q6 Focusing on class Hospital and its relationships, which of the following entries is true?:

- ☐ A doctor may work for several hospitals
- ☐ A hospital may have no laboratories
- ☐ Each hospital has at least a doctor

Q7 Focusing on classes Examination, Patient, and Inspection, which of the following entries is true?:

- ☐ Given a patient, it is possible to obtain the diagnosis for an inspection regarding an examination
- ☐ It is possible to obtain the reservation date of an inspection for a given examination.
- ☐ A patient makes at least one inspection

Q8 Focusing on class Person and its specialization which of the following entries is true?:

- ☐ A person has a hiring date
- ☐ It is possible to obtain the registration date of a person who made an examination
- ☐ A doctor may require an examination

Q9 Focusing on classes Laboratory and Hospital, which of the following entries is true?:

- ☐ A hospital has one or more laboratories
- ☐ A laboratory may be identified by its number
- ☐ It is possible to obtain the floor of a laboratory in a given hospital.

Q10 Focusing on the relationship between Doctor and Hospital, which of the following entries is true?:

- ☐ A doctor may work for no hospital
- ☐ A doctor works for an hospital and several doctors collaborate for an Hospital
- ☐ A doctor works at least for an hospital and at least a doctor works for a hospital

End time:

Final questionnaire

The following questions have to answered after the execution of the comprehension tasks

A1 I had enough time to perform the tasks

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree

A2. The task objectives were perfectly clear to me.

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree

A3. The tasks I performed were perfectly clear to me.

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree

A4. Judge the difficulty of the comprehension task

- ☐ Strongly disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly agree